Human Body Systems: Breathing & Transport

Test · 38 marks total

Name: ________________________________	Date: ______________	Score: _____ / 38

Section A — Respiration

A1. Complete the word equation for aerobic respiration: glucose + oxygen → ______ + ______.

[2 marks]

A2. State two differences between aerobic and anaerobic respiration in muscle cells.

[2 marks]

A3. During intense exercise, a sprinter's muscles cannot get enough oxygen. Name the type of respiration used, and describe one problem this causes.

[2 marks]

Section B — Exchange surfaces

B1. State three features that make a surface efficient for exchanging substances such as gases.

[3 marks]

B2. Explain how the structure of alveoli is adapted for efficient gas exchange.

[4 marks]

B3. Explain why capillary walls are only one cell thick.

[2 marks]

Section C — The heart & circulation

C1. Name the four chambers of the heart.

[4 marks]

C2. What is meant by 'double circulation'?

[2 marks]

C3. Explain why the wall of the left ventricle is thicker (more muscular) than the wall of the right ventricle.

[2 marks]

Section D — Blood vessels

D1. State one structural difference between an artery and a vein, and explain why this difference is needed.

[2 marks]

D2. Veins contain valves. Explain why these are needed.

[2 marks]

Section E — Blood components

E1. Name the four components of blood.

[4 marks]

E2. Explain how red blood cells are adapted to carry oxygen efficiently.

[3 marks]

E3. What is the function of platelets?

[1 mark]

Section F — Apply it

F1. A patient has a low red blood cell count, a condition called anaemia. Suggest why they might feel tired all the time.

[3 marks]